

Pop-Quiz

- Derivatives wrt support points: $\nabla_{\mathbf{x}_i} \text{OT}_c^\varepsilon(\mathbf{a}, \mathbf{b}) = \sum_{j=1}^m \pi_{i,j}^* \nabla_x c(\mathbf{x}_i, \mathbf{y}_j)$
- Show that for $c(x, y) = \|x - y\|^2$, $\nabla_{\mathbf{x}_i} \text{OT}_c^\varepsilon(\mathbf{a}, \mathbf{b}) \propto \text{Id} - T_{\text{bary}}$
- Solution:

$$\begin{aligned} \nabla_{\mathbf{x}_i} \text{OT}_c^\varepsilon(\mathbf{a}, \mathbf{b}) &= \sum_{j=1}^m \pi_{i,j} \nabla_x c(\mathbf{x}_i, \mathbf{y}_j) = \sum_{j=1}^m \pi_{i,j} 2(\mathbf{x}_i - \mathbf{y}_j) \\ &= 2\left(\mathbf{a}_i \mathbf{x}_i - \sum_{j=1}^m \pi_{i,j} \mathbf{y}_j\right) = 2\mathbf{a}_i \left(\mathbf{x}_i - \frac{1}{\mathbf{a}_i} \sum_{j=1}^m \pi_{i,j} \mathbf{y}_j\right)(\mathbf{x}_i) = \lambda(\text{Id} - T_{\text{bary}})(\mathbf{x}_i) \end{aligned}$$

Differentiability of EOT: General Case